

6(a)	$^{147}_{62}\text{Sm}$	A1
6(b)(i)	the (kinetic) energy of the particles is discrete / has only one value (so must be alpha)	B1
	and beta particles have a (continuous) range of (kinetic) energies (so can't be beta)	B1
6(b)(ii)	$^{147}_{62}\text{Sm} \rightarrow ^{143}_{60}\text{Nd} + ^4_2\alpha$ values for Sm and Nd correct with no other extra particles on either side of the equation	A1
	$^4_2\alpha$ correct	A1
6(c)(i)	up quark charge is $+(2/3)$ (e) or bottom quark charge is $-(1/3)$ (e)	C1
	$0 = +(2/3) (e) - (1/3) (e) + q$	A1
	(so) charge (on third quark must be) $-(1/3) (e)$	
6(c)(ii)	down or strange	A1